

Factorising & Pythagoras

Hoofstuk 8 (Faktoriserings) • Hoofstuk 13 (Pythagoras) • Exam Focus p164 & p239

Everything you need to pass tomorrow's test. Every rule, every method, and a fully worked solution to every question on the worksheet.

Work through Part A → Part B → Part C, then do the 12-question self-test at the end. Answers are on the last page.

How to use this in one evening

Time	What to do
0–10 min	Read The Golden Order (page 2) until you can say the 4 steps from memory. This single page decides most of your marks.
10–40 min	Part A: the four factorising rules. Cover the answer, try each worked example yourself first, then check.
40–55 min	Part B: algebraic fractions (factorise top, factorise bottom, cancel <i>brackets</i>).
55–80 min	Part C: Pythagoras — the theorem, the converse, and the acute/obtuse test.
80–95 min	Read Top 10 mistakes , then do the self-test. Anything you get wrong, re-read that rule.

THE ONE-SENTENCE SUMMARY

Chapter 8: always take out the common factor first, then look at how many terms are left — 2 terms → difference of squares, 3 terms → trinomial, 4 terms → group in pairs.

Chapter 13: $a^2 + b^2 = c^2$ where c is always the longest side (the hypotenuse, opposite the right angle). To *find* the hypotenuse you **add**; to find a shorter side you **subtract**.

Bilingual glossary — Afrikaans / English

Faktoriseer volledig	Factorise fully (keep going until nothing factorises further)
Vereenvoudig	Simplify
Gemeenskaplike faktor	Common factor
Verskil van twee kwadrate	Difference of two squares
Drieterm / Tweeterm	Trinomial (3 terms) / Binomial (2 terms)
Wortelvorm	Surd form (leave the $\sqrt{\quad}$ sign, do not round off)
Reghoekige driehoek	Right-angled triangle
Skuinssy	Hypotenuse (longest side)
Regte hoek	Right angle (90°)
Skerphoekig / Stomphoekig	Acute-angled / Obtuse-angled

Vierkant / Reghoek	Square / Rectangle
Bereken die lengte van	Calculate the length of
Toon aan dat	Show that (you must show the working, not just the answer)
Korrek tot twee desimale syfers	Correct to two decimal places

A1. The Golden Order — do these 4 checks, in this order, every time

Factorising is not guessing. It is a **decision tree**. Ask the questions in order and stop when the expression is fully broken down.

1. Common factor?

Take out the HCF of every term — numbers *and* letters.
Always do this first, always.

2. Two terms?

Difference of two squares:
 $a^2 - b^2 = (a-b)(a+b)$

3. Three terms?

Trinomial: find two numbers that $\times$ to give the last term and $+$ to give the middle.

4. Four terms?

Group in pairs, take out a common factor from each pair, then take out the common bracket.

"FACTORISE FULLY" MEANS KEEP GOING

After you factorise once, **look at each bracket again** and run the checklist on it. Marks are lost every year for stopping at $8(x^4-1)$ when x^4-1 still factorises. If a bracket has 2 terms and both are perfect squares, it factorises again.

A2. Rule 1 — Common factor (gemeenskaplike faktor)

Find the **highest common factor** of all terms and write it outside a bracket. Divide each original term by the HCF to get what goes inside.

RULE

$ka + kb = k(a + b)$ — for the letters, take the **lowest power** that appears in every term.

Worked — (d)(1) $7x + 14$

HCF of 7 and 14 is 7. There is no x in the second term, so no letter comes out.

$7x + 14 = 7(x + 2)$ — check: $7 \times x = 7x$ ✓, $7 \times 2 = 14$ ✓

$$= 7(x + 2)$$

Worked — (d)(7) $4a^2 + 3a$

Numbers: HCF of 4 and 3 is 1. Letters: a^2 and $a \rightarrow$ lowest power is a .

So the HCF is a . $4a^2 \div a = 4a$ and $3a \div a = 3$.

$$= a(4a + 3)$$

WATCH OUT

Don't write $a(4a + 3a)$. You divide each term by the HCF — you do not just copy it. Multiply back out to check, every time.

A3. Rule 2 — Difference of two squares (verskil van twee kwadrate)

RULE

$$a^2 - b^2 = (a - b)(a + b)$$

Three conditions, all must be true: (i) exactly two terms, (ii) a **minus** between them, (iii) both terms are perfect squares.

$a^2 + b^2$ does NOT factorise. A sum of two squares is prime. This is a favourite trap question.

Square roots you must know: $\sqrt{4m^2} = 2m$, $\sqrt{9x^2} = 3x$, $\sqrt{25} = 5$, $\sqrt{x^8} = x^4$ (halve the power), $\sqrt{16y^6} = 4y^3$.

Worked — (d)(10) $4m^2 - 1$

Two terms, minus, both perfect squares: $\sqrt{4m^2} = 2m$, $\sqrt{1} = 1$.

$$= (2m - 1)(2m + 1)$$

Worked — (d)(21) $8x^8 - 8$ (the "factorise fully" trap)

Step 1 — common factor first: HCF is 8. $= 8(x^8 - 1)$

Step 2 — the bracket has 2 terms, minus, both squares: $= 8(x^4 - 1)(x^4 + 1)$

Step 3 — look again! $x^4 - 1$ is *still* a difference of squares: $= 8(x^2 - 1)(x^2 + 1)(x^4 + 1)$

Step 4 — look again! $x^2 - 1$ is *still* a difference of squares:

$$= 8(x - 1)(x + 1)(x^2 + 1)(x^4 + 1)$$

Stop here: $x^2 + 1$ and $x^4 + 1$ are SUMS of squares — they do not factorise.

A4. Rule 3 — Trinomials $x^2 + bx + c$ (drieterme)

METHOD

Find two numbers that **multiply to give c** (the last number) and **add to give b** (the middle number). Then

$$x^2 + bx + c = (x + \text{first})(x + \text{second}).$$

The sign table — learn this, it removes all the guessing

Last sign (c)	Middle sign (b)	Signs in the brackets	What the two numbers do	Example
+	+	$(+)(+)$ — both plus	ADD to b , multiply to c	$x^2 + 12x + 27 = (x + 3)(x + 9)$
+	-	$(-)(-)$ — both minus	ADD to b , multiply to c	$k^2 - 12k + 27 = (k - 3)(k - 9)$
-	+	$(+)(-)$ — one of each	SUBTRACT to b , the bigger number is +	$a^2 + 6a - 27 = (a + 9)(a - 3)$
-	-	$(+)(-)$ — one of each	SUBTRACT to b , the bigger number is -	$y^2 - 6y - 27 = (y - 9)(y + 3)$

SHORTCUT TO REMEMBER IT

Last sign PLUS → both brackets get the **middle sign**, and the numbers **add up** to the middle number.

Last sign MINUS → the brackets get **one plus and one minus**, the numbers have a **difference** equal to the middle

number, and the **bigger number takes the middle sign**.

Factor pairs of 27 (used in all four questions below): 1×27 and 3×9 . Since we need 12 or 6 from them: $3 + 9 = 12$ and $9 - 3 = 6$.

(d)(22) $x^2 + 12x + 27$

Last +, middle + $\rightarrow$ both brackets +.

Need: $\times = 27$, $+$ = 12 $\rightarrow$ **3 and 9**.

$$= (x + 3)(x + 9)$$

(d)(23) $k^2 - 12k + 27$

Last +, middle - $\rightarrow$ both brackets -.

Need: $\times = 27$, $+$ = 12 $\rightarrow$ **3 and 9**.

$$= (k - 3)(k - 9)$$

(d)(24) $a^2 + 6a - 27$

Last - $\rightarrow$ one +, one -.

Need: $\times = 27$, difference = 6 $\rightarrow$ **9 and 3**.

Middle is +, so the bigger (9) is +.

$$= (a + 9)(a - 3)$$

(d)(25) $y^2 - 6y - 27$

Last - $\rightarrow$ one +, one -.

Need: $\times = 27$, difference = 6 $\rightarrow$ **9 and 3**.

Middle is -, so the bigger (9) is -.

$$= (y - 9)(y + 3)$$

ALWAYS CHECK BY MULTIPLYING BACK

FOIL $(y - 9)(y + 3) = y^2 + 3y - 9y - 27 = y^2 - 6y - 27 \checkmark$. This takes 5 seconds and catches every sign error.

A5. Rule 4 — Common bracket & grouping (four terms)

RULE 4A — A BRACKET CAN BE A COMMON FACTOR

If the same bracket appears in every term, take the whole bracket out:

$$4x(a+b) + 3(a+b) = (a+b)(4x + 3)$$

Worked — (e)(1) $4x(a + b) + 3(a + b)$

The bracket $(a + b)$ is in both terms $\rightarrow$ it is the common factor.

What is left over: $4x$ from the first term, $+3$ from the second.

$$= (a + b)(4x + 3)$$

RULE 4B — THE SIGN-FLIP TRICK (VERY IMPORTANT)

$$(1 - x) = -(x - 1) \quad (-4 - x) = -(x + 4) \quad (b - a) = -(a - b)$$

If two brackets are the **same but backwards**, take a -1 out of one of them to make them match. Then you can factorise.

Worked — (e)(4) $x^2(x - 1) + 4(1 - x)$

The brackets are backwards: $(x - 1)$ vs $(1 - x)$.

Flip the second: $4(1 - x) = -4(x - 1)$

$$= x^2(x - 1) - 4(x - 1) \quad \text{— now the brackets match}$$

Take out $(x - 1)$: $= (x - 1)(x^2 - 4)$

Factorise fully! $x^2 - 4$ is a difference of two squares.

$$= (x - 1)(x - 2)(x + 2)$$

Worked — Page 2, (5) $x^2(x + 4) + 9(-4 - x)$

$$(-4 - x) = -(4 + x) = -(x + 4) \quad \text{— take out } -1, \text{ then reorder}$$

$$= x^2(x + 4) - 9(x + 4)$$

$$= (x + 4)(x^2 - 9) \quad \text{— difference of squares in the bracket}$$

$$= (x + 4)(x - 3)(x + 3)$$

RULE 4C — GROUPING FOUR TERMS IN PAIRS

Step 1: split into two pairs. **Step 2:** take a common factor out of each pair. **Step 3:** the two brackets left must be identical — take that bracket out.

If the brackets are *not* identical, either re-pair the terms differently, or take out a negative from the second pair.

Worked — (e)(7) $ax + 3a + bx + 3b$

Pair them: $(ax + 3a) + (bx + 3b)$

Common factor of pair 1 is a ; of pair 2 is b :

$$= a(x + 3) + b(x + 3) \quad \text{— the brackets match } \checkmark$$

Take out $(x + 3)$:

$$= (x + 3)(a + b)$$

IF THE PAIRS DON'T MATCH

Try the other pairing: $(ax + bx) + (3a + 3b) = x(a + b) + 3(a + b) = (a + b)(x + 3)$ — the same answer. Order of brackets doesn't matter.

A6. Chapter 8 — complete answer list

No.	Question	Answer	Rule used
(d)1	$7x + 14$	$7(x + 2)$	Common factor
(d)7	$4a^2 + 3a$	$a(4a + 3)$	Common factor (letters only)
(d)10	$4m^2 - 1$	$(2m - 1)(2m + 1)$	Difference of squares
(d)21	$8x^3 - 8$	$8(x - 1)(x + 1)(x^2 + 1)(x^4 + 1)$	Common factor, then diff. of squares $\times 3$
(d)22	$x^2 + 12x + 27$	$(x + 3)(x + 9)$	Trinomial (+ +)
(d)23	$k^2 - 12k + 27$	$(k - 3)(k - 9)$	Trinomial (- -)
(d)24	$a^2 + 6a - 27$	$(a + 9)(a - 3)$	Trinomial (+ -), bigger is +
(d)25	$y^2 - 6y - 27$	$(y - 9)(y + 3)$	Trinomial (+ -), bigger is -
(e)1	$4x(a + b) + 3(a + b)$	$(a + b)(4x + 3)$	Common bracket
(e)4	$x^2(x - 1) + 4(1 - x)$	$(x - 1)(x - 2)(x + 2)$	Sign flip $\rightarrow$ common bracket $\rightarrow$ diff. squares
(e)7	$ax + 3a + bx + 3b$	$(x + 3)(a + b)$	Grouping in pairs
p2 (5)	$x^2(x + 4) + 9(-4 - x)$	$(x + 4)(x - 3)(x + 3)$	Sign flip $\rightarrow$ common bracket $\rightarrow$ diff. squares

B1. The method — three steps, never anything else

METHOD

1. **Factorise the top** (numerator) completely.
2. **Factorise the bottom** (denominator) completely.
3. **Cancel whole brackets / whole factors** that appear top and bottom.

THE GOLDEN RULE OF CANCELLING

You may **only cancel factors** (things being **multiplied**). You may **never cancel terms** (things being **added or subtracted**).

$$\frac{x+3}{x} \text{ — you cannot cancel the } x\text{'s.} \quad \frac{(x+3)(x-1)}{(x-1)} = x+3 \text{ — you can cancel the bracket. } \checkmark$$

Worked — (i)(2) $\frac{x^3y - 4xy^2}{x^2y^2}$

Top: common factor of x^3y and $4xy^2$ is $xy \rightarrow xy(x^2 - 4y)$

Bottom: $x^2y^2 = xy \cdot xy$

$$\frac{xy(x^2 - 4y)}{x^2y^2} \rightarrow \text{cancel one } x \text{ and one } y$$

$$= \frac{x^2 - 4y}{xy}$$

Do NOT cancel further — $x^2 - 4y$ is a subtraction, not a product.

Worked — (i)(3) $\frac{x^2 - 9}{x^2 - 3x}$

Top: difference of squares $\rightarrow (x-3)(x+3)$

Bottom: common factor $x \rightarrow x(x-3)$

$$\frac{(x-3)(x+3)}{x(x-3)} \rightarrow \text{cancel } (x-3)$$

$$= \frac{x+3}{x}$$

Worked — (i)(4) $\frac{x^2 - 14x + 24}{x^2 - 11x - 12}$

Top: $\times = 24$, $+$ $= 14$, both signs $- \rightarrow 2$ and $12 \rightarrow (x-2)(x-12)$

Bottom: last is $- \rightarrow$ one $+$ one $-$; $\times = 12$, difference $= 11 \rightarrow 12$ and 1 ; middle is $-$ so bigger is $- \rightarrow (x-12)(x+1)$

$$\frac{(x-2)(x-12)}{(x-12)(x+1)} \rightarrow \text{cancel } (x-12)$$

$$= \frac{x-2}{x+1}$$

Worked — (i)(5) $\frac{x^2 - 14x}{x^2 - 16x + 28}$

Top: common factor $x \rightarrow x(x - 14)$

Bottom: $\times = 28$, $+= 16$, both $- \rightarrow 2$ and $14 \rightarrow (x - 14)(x - 2)$

$$\frac{x(x-14)}{(x-14)(x-2)} \rightarrow \text{cancel } (x-14)$$

$$= \frac{x}{x-2}$$

EXAM HABIT

If the top and bottom don't share a bracket after factorising, you have probably made a sign mistake in one of the trinomials — go back and check. These questions are *designed* to cancel.

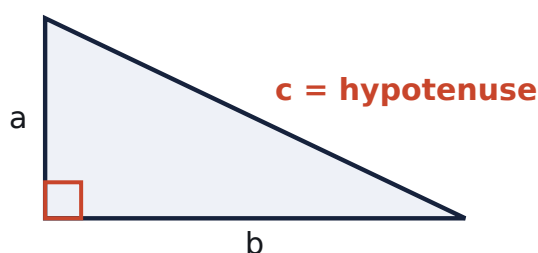
C1. The Theorem of Pythagoras

THE THEOREM

In a right-angled triangle, **the square on the hypotenuse equals the sum of the squares on the other two sides**:

$$a^2 + b^2 = c^2$$

where c is the **hypotenuse** — the side **opposite the right angle**, and always the **longest** side.



WHICH OPERATION? ADD OR SUBTRACT?

What you are looking for	What you do	Formula
The hypotenuse (longest side, opposite 90°)	ADD the two squares, then $\sqrt{}$	$c = \sqrt{a^2 + b^2}$
A shorter side (a leg)	SUBTRACT the smaller square from the hypotenuse squared, then $\sqrt{}$	$a = \sqrt{c^2 - b^2}$

THE #1 PYTHAGORAS MISTAKE

Adding when you should subtract. **Before you calculate, ask: is the side I want the longest side of that triangle?** If yes → add. If no → subtract. A shorter side can never come out bigger than the hypotenuse — if it does, you added by mistake.

Surd form vs decimals (wortelvorm)

- "Leave your answer in surd form" / "gee jou antwoord in wortelvorm" → keep the $\sqrt{}$ sign, do not round. Simplify it if you can: $\sqrt{98} = \sqrt{49 \times 2} = 7\sqrt{2}$.
- "Correct to two decimal places" → use the calculator and round: $\sqrt{87} = 9.327... \approx 9.33$.
- To simplify a surd, split off the largest **perfect square** factor: $\sqrt{50} = \sqrt{25 \cdot 2} = 5\sqrt{2}$, $\sqrt{72} = \sqrt{36 \cdot 2} = 6\sqrt{2}$, $\sqrt{12} = 2\sqrt{3}$, $\sqrt{98} = 7\sqrt{2}$.

Perfect squares & Pythagorean triples worth memorising

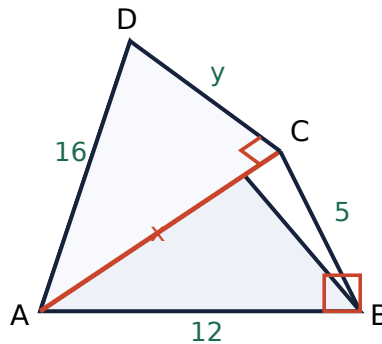
Squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225, 256, 289, 324, 361, 400.

Triples (and their multiples) — spotting these saves time: **3,4,5 · 5,12,13 · 8,15,17 · 7,24,25 · 20,21,29**.

Multiples count too: $(5,12,13) \times 4 = (20,48,52)$ and $(20,21,29) \times 4 = (80,84,116)$ — both appear on this worksheet!

C2. Two-triangle problems — Exercise 13.1 (d)(1)

The rule: when two right-angled triangles share a side, **solve the triangle where you know two sides first**, then use that answer in the second triangle.



Right angles at **B** (in $\triangle ABC$) and at **C** (in $\triangle ADC$). $AB = 12$, $BC = 5$, $AD = 16$.

Find $x (= AC)$ — use $\triangle ABC$

Right angle is at B, so **AC is the hypotenuse** → **ADD**.

$$AC^2 = AB^2 + BC^2$$

$$x^2 = 12^2 + 5^2 = 144 + 25 = 169$$

$$x = \sqrt{169} \quad \mathbf{x = 13} \quad (\text{the 5-12-13 triple})$$

Find $y (= DC)$ — use $\triangle ADC$

Right angle is at C, so **AD (16) is the hypotenuse**, and DC is a shorter side → **SUBTRACT**.

$$AD^2 = DC^2 + AC^2 \rightarrow 16^2 = y^2 + 13^2$$

$$y^2 = 256 - 169 = 87$$

$$y = \sqrt{87} \quad \mathbf{y \approx 9.33} \quad (2 \text{ d.p.}; \text{ in surd form } \sqrt{87} \text{ — } 87 = 3 \times 29 \text{ has no square factor, so it cannot simplify})$$

NOTICE

You could not find y first — $\triangle ADC$ only had one known side until x gave you the second. **Always look for the triangle with two known sides.**

C3. The CONVERSE of Pythagoras — proving a right angle

CONVERSE

If the square on the longest side **equals** the sum of the squares on the other two sides, then the triangle is **right-angled**, and the right angle is **opposite the longest side**.

HOW TO LAY OUT A "SHOW THAT" ANSWER (THIS IS HOW MARKS ARE AWARDED)

1. Identify the **longest side**.
2. Work out **LHS** = (longest side)².
3. Work out **RHS** = (short side)² + (other short side)².
4. Write **LHS = RHS**.
5. Conclude: $\therefore \triangle ABC$ is **right-angled** (converse of Pythagoras), and $\therefore$ **the right angle is at** ____ — the vertex **opposite the longest side**.

13.2 (a)(1) — In $\triangle ABC$: $AB = 20$, $BC = 48$, $AC = 52$

Longest side = **AC = 52**.

LHS: $AC^2 = 52^2 = 2704$

RHS: $AB^2 + BC^2 = 20^2 + 48^2 = 400 + 2304 = 2704$

$\therefore AC^2 = AB^2 + BC^2$

$\therefore \triangle ABC$ is right-angled, with the right angle at B — B is opposite AC (converse of Pythagoras)

Sketch: a triangle with a square-corner mark at B, 20 and 48 on the sides meeting at B, 52 across from B.

13.2 (a)(2) — In $\triangle ABC$: $c = 17$, $b = 8$, $a = 15$

Small letters name the side **opposite** the matching capital: a is opposite A, b opposite B, c opposite C.

Longest side = **c = 17**.

LHS: $c^2 = 17^2 = 289$

RHS: $a^2 + b^2 = 15^2 + 8^2 = 225 + 64 = 289$

$\therefore c^2 = a^2 + b^2$

$\therefore \triangle ABC$ is right-angled, with the right angle at C — C is opposite side c, the longest side (the 8–15–17 triple)

C4. Acute-angled or obtuse-angled? (skerphoekig of stomphoekig)

THE TEST — COMPARE (LONGEST SIDE)² WITH THE SUM OF THE OTHER TWO SQUARES

If...	Then the triangle is...	Where is the special angle?
$c^2 = a^2 + b^2$	Right-angled (reghoekig)	90° opposite the longest side
$c^2 < a^2 + b^2$	Acute-angled (skerphoekig) — all angles < 90°	—

$$c^2 > a^2 + b^2$$

Obtuse-angled (stomphoekig)

The obtuse angle is **opposite the longest side**

HOW TO REMEMBER

The **bigger** the longest side is compared to the other two, the more it is "stretched open" — so **bigger** $c^2 \rightarrow$ **bigger (obtuse) angle**. Small $c^2 \rightarrow$ the angle closes up $\rightarrow$ acute.

13.2 (b)(1) — In $\triangle ABC$: $AB = 24$, $BC = 14$, $AC = 27$

Longest side = $AC = 27$. It is opposite **B**.

$$AC^2 = 27^2 = 729$$

$$AB^2 + BC^2 = 24^2 + 14^2 = 576 + 196 = 772$$

$$729 < 772, \text{ so } AC^2 < AB^2 + BC^2$$

$\therefore \triangle ABC$ is **acute-angled** (all three angles are less than 90° , so there is no obtuse angle to name)

13.2 (b)(2) — In $\triangle DEF$: $d = 8$, $e = 18$, $f = 13$

Longest side = $e = 18$. Side e is opposite **E**.

$$e^2 = 18^2 = 324$$

$$d^2 + f^2 = 8^2 + 13^2 = 64 + 169 = 233$$

$$324 > 233, \text{ so } e^2 > d^2 + f^2$$

$\therefore \triangle DEF$ is **obtuse-angled, and the obtuse angle is $\hat{E}$** (the angle at E, opposite the longest side)

C5. Exam Focus p239 — squares and rectangles

FACTS YOU MUST KNOW ABOUT SQUARES AND RECTANGLES

- All four angles are 90° , so a **diagonal always creates a right-angled triangle** — that is your Pythagoras triangle.
- In a **square**, all four sides are equal. Diagonal of a square of side s is $s\sqrt{2}$.
- In a **rectangle**, opposite sides are equal.
- The diagonals of a rectangle are equal AND they bisect each other** (cut each other in half). So if they meet at E:
 $AE = EC = BE = ED = \frac{1}{2} \times \text{diagonal}$.

(b) ABCD is a square with side 7 cm. Calculate BD, in surd form.

BD is a diagonal $\rightarrow$ it makes right-angled $\triangle ABD$ (right angle at A).

All sides of a square are equal, so $AB = AD = 7$ cm.

BD is opposite the right angle $\rightarrow$ BD is the hypotenuse $\rightarrow$ **ADD**.

$$BD^2 = AB^2 + AD^2 = 7^2 + 7^2 = 49 + 49 = 98$$

$$BD = \sqrt{98} = \sqrt{(49 \times 2)} = \sqrt{49} \times \sqrt{2}$$

$$\boxed{BD = 7\sqrt{2} \text{ cm}} \quad \text{— surd form: do NOT write 9.90}$$

(c) ABCD is a rectangle, AD = 84 and AB = 80. Calculate AE.

E is where the diagonals cross, so $AE = \frac{1}{2} AC$. **First find the whole diagonal AC.**

$\triangle ABC$ has a right angle at B, with $AB = 80$ and $BC = AD = 84$ (opposite sides of a rectangle are equal).

AC is the hypotenuse $\rightarrow$ **ADD**:

$$AC^2 = 80^2 + 84^2 = 6400 + 7056 = 13456$$

$$AC = \sqrt{13456} = 116 \quad (\text{it's the 20–21–29 triple} \times 4)$$

The diagonals of a rectangle bisect each other:

$$AE = \frac{1}{2} \times 116 \quad \boxed{AE = 58}$$

DON'T STOP EARLY

The question asks for **AE**, not AC. Finding 116 and writing that down loses the last mark. **Underline what the question actually asks for before you start.**

Top 10 mistakes to avoid tomorrow

#	The mistake	The fix
1	Not taking out the common factor first.	Common factor is always check number 1.
2	Stopping too early on "factorise fully " — e.g. leaving $8(x^4 - 1)$.	Re-check every bracket after each step.
3	Trying to factorise $a^2 + b^2$ (a sum of squares).	It doesn't factorise. Only the minus version does.
4	Sign errors in trinomials.	Use the sign table, then FOIL back to check.

5	Not spotting $(1 - x) = -(x - 1)$.	Backwards brackets → take out -1 to match them.
6	Cancelling terms in a fraction instead of factors .	Cancel only complete brackets/products.
7	Adding in Pythagoras when you should subtract.	Ask first: "is the side I want the hypotenuse?"
8	Forgetting to take the square root at the end.	You found x^2 , not x . Finish the job.
9	Rounding when it says surd form (or vice versa).	"Wortelvorm" = keep $\sqrt{}$. "2 decimal places" = round.
10	Answering the wrong quantity (AC instead of AE).	Underline what is asked before starting.

Self-test — 12 questions, 20 minutes, no notes

Cover the answers at the bottom. Aim for 10/12. Anything you miss, go straight back to that rule.

1. Factorise: $5x + 20$
2. Factorise: $6p^2 + 9p$
3. Factorise fully: $3x^2 - 27$
4. Factorise: $x^2 + 10x + 21$
5. Factorise: $m^2 - 4m - 21$
6. Factorise: $t^2 - 10t + 21$
7. Factorise: $5a(x - 2) - 3(2 - x)$
8. Factorise: $pq + 4p + 5q + 20$
9. Simplify: $(x^2 - 25) \div (x^2 + 5x)$
0. A right-angled triangle has legs 9 and 12. Find the hypotenuse.
1. In $\triangle PQR$, $PQ = 9$, $QR = 40$, $PR = 41$. Show it is right-angled and state which angle is 90° .
2. A square has side 5 cm. Find its diagonal in surd form.

Answers

1. $5(x + 4)$	2. $3p(2p + 3)$	3. $3(x - 3)(x + 3)$
4. $(x + 3)(x + 7)$	5. $(m - 7)(m + 3)$	6. $(t - 3)(t - 7)$
7. $(x - 2)(5a + 3) [-3(2 - x) = +3(x - 2)]$	8. $(q + 4)(p + 5)$	9. $(x - 5)/x$
10. 15 $[3 \cdot 4 \cdot 5 \times 3]$	11. $81 + 1600 = 1681 = 41^2 \therefore$ right-angled, 90° at Q	12. $5\sqrt{2}$ cm

FINAL 60 SECONDS BEFORE THE TEST

Factorising: Common factor $\rightarrow$ 2 terms (difference of squares) $\rightarrow$ 3 terms (trinomial + sign table) $\rightarrow$ 4 terms (grouping). Then **check every bracket again**.

Fractions: Factorise top, factorise bottom, cancel **brackets** only.

Pythagoras: $a^2 + b^2 = c^2$, c = longest = opposite 90° . Hypotenuse $\rightarrow$ ADD. Shorter side $\rightarrow$ SUBTRACT. Then $\sqrt{\quad}$.

Converse: longest² = sum of other two squares $\rightarrow$ right-angled, 90° opposite the longest side. Less than $\rightarrow$ acute. Greater than $\rightarrow$ obtuse (obtuse angle opposite the longest side).

Show every step. Method marks are real marks. Good luck! 🍀